



ROYAL UNIVERSITY OF BHUTAN
COLLEGE OF SCIENCE AND TECHNOLOGY
RINCHENDING; PHUENTSHOLING



CTE204
Web Application Development
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Conceptual Notes

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1. Originality of the Concept

The proposed application is called **Smart Study Session Planner**. It is a web-based study management application that helps students create effective and personalized study sessions based on their available time, energy level, and subject difficulty. Unlike a normal study timer or to-do list, this application includes intelligent study interval generation, session tracking, note-taking, and automatic review question generation from the user's notes.

I chose this idea because students often struggle to manage study time efficiently, especially when handling multiple difficult subjects. Many existing study applications only provide timers or schedules, but this application attempts to make study sessions smarter and more interactive by adapting to the student's condition and helping with revision preparation.

This application differs from generic examples because it combines:

- Study session planning
- Smart focus interval calculation
- Notes management
- Auto-generated review questions
- Session statistics and performance tracking

The application is designed specifically for students and focuses on productivity and active learning rather than simple task management.

2. Relevancy

The target users of this application are college students and high school students who want to improve their study habits and organize their learning sessions effectively.

Students frequently face problems such as:

- Poor time management
- Difficulty concentrating during long study sessions
- Lack of revision preparation
- Forgetting important concepts after studying
- Inconsistent study schedules

The Smart Study Session Planner helps solve these problems by allowing users to:

- Create structured study sessions
- Adjust sessions according to difficulty and energy level

- Take notes during study sessions
- Generate review questions automatically for self-testing
- Monitor completed sessions and study effectiveness

This application is useful because it encourages active learning, improves organization, and helps students maintain productive study routines.

3. Clarity of the Proposal

The application will include the following features:

Features	Description
Study Session Creation	Users can create a study session by entering subject name, difficulty level, energy level, and available study duration.
Smart Focus Interval Generator	The application automatically calculates recommended study intervals based on the difficulty level of the subject.
Dynamic Range Sliders	Users can interact with sliders to set difficulty and energy levels with real-time value updates.
Form Validation	The system validates user inputs such as subject name length and duration range before creating a session.
Session Notes Area	Users can write study notes during active study sessions.
Automatic Review Question Generator	The system extracts important keywords from notes and generates review questions for revision purposes.
Study Session Tracking	Users can view all created study sessions and monitor whether sessions are completed or pending.
Session Statistics Dashboard	The application displays statistics such as completed sessions and average study effectiveness.
Local Storage Support	All study sessions are saved using browser local storage so data remains available after refreshing the page.
Responsive User Interface	The application interface is responsive and can adapt to mobile devices, tablets, and desktop screens.

Conclusion

The Smart Study Session Planner is a productivity-focused web application designed to help students manage study sessions more effectively. By combining planning, note-taking, automatic question generation, and performance tracking, the application provides a smarter and more interactive study experience for students.